

Attention Mechanisms for Landmark-Based Turkish Sign Language Recognition

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Abstract

Isolated sign language recognition from video is a challenging sequence classification problem: informative motion cues can occur at any point in a clip, and models trained without care can silently learn signer-specific habits instead of the sign itself. This project builds a landmark-based recognition pipeline for a 20-word subset of the Turkish Sign Language AUTSL dataset [4]. Each video is converted into a 450-dimensional per-frame feature vector: 225 raw landmark coordinates (33 pose and 42 hand landmarks) obtained with MediaPipe's Pose and Hand Landmarker models [5], concatenated with a matching set of inter-frame velocity features. Three LSTM-based classifiers are compared: a baseline that uses only the final hidden state, a temporal Attention-LSTM that learns to weight informative frames [2], and a novel Spatial+Temporal Attention-LSTM that additionally learns which body region (pose, left hand, right hand) is most informative at each frame. All models are evaluated with a signer-independent train/validation/test split, which is shown empirically to give a more realistic (and lower) accuracy estimate than a naive random split. On a held-out test set of 246 signer-independent examples, a single training run of the temporal Attention-LSTM reaches 60.6% accuracy (59.4% macro-F1) against a 53.3% baseline and a 33.7% Spatial+Temporal model, with a two-model (Baseline+Attention) ensemble reaching only 55.7% – below the Attention-LSTM alone. Repeating training with three different random seeds, however, shows that this single-run advantage is not reliable: the Baseline LSTM's 3-seed mean accuracy ($51.6\% \pm 3.9$) is in fact marginally *higher* than the Attention-LSTM's ($49.9\% \pm 1.6$), with heavily overlapping variance ranges, though the Attention-LSTM is consistently more stable across seeds (roughly half the standard deviation). The Spatial+Temporal model, in contrast, is the one architecture whose weakness is robust across both single-run and 3-seed evaluation, with no overlap against either other model, and its confusion matrix reveals a mode-collapse pattern that is analyzed in detail. This project's central methodological finding is therefore that single-run comparisons between closely-matched architectures in this low-resource setting are unreliable and can reverse under multi-seed re-evaluation – an important caveat for small sign-language datasets.

Keywords: sign language recognition; attention mechanism; recurrent neural networks; landmark-based action recognition; low-resource learning.

1 Introduction

Sign languages are full natural languages, but automatic recognition systems for them remain far less mature than systems for spoken language. Isolated sign language recognition (ISLR) – classifying a short video clip into one of a fixed vocabulary of signs – is a natural first step toward larger goals such as continuous sign language translation, and it exposes the same core difficulty that continuous systems face: the information that identifies a sign is distributed unevenly across a clip's frames and across the signer's body (dominant hand, non-dominant hand, torso posture), rather than concentrated at a fixed position.

This project addresses ISLR on a 20-word subset of the Turkish Sign Language AUTSL dataset [4], using body and hand landmark sequences rather than raw pixels as input. Working with landmarks instead of RGB frames removes background, lighting, and clothing as nuisance variables, and keeps the model's input small enough to be trained from scratch on a modest, single-GPU budget – a realistic constraint for a course project.

The main limitation of a naive recurrent baseline for this task is that it is forced to compress an entire 30-frame clip into a single hidden state, discarding the fact that different signs become distinguishable at different points in the motion. A second, less discussed limitation is methodological: many small sign-language datasets contain multiple clips per signer, and a randomly chosen train/test split can let the same signer's clips appear in both sets, silently inflating reported accuracy by rewarding the model for recognizing the person rather than the sign.

This project's contribution is threefold:

- An empirical comparison of three LSTM-based architectures – a last-hidden-state baseline, a temporal Attention-LSTM, and a novel Spatial+Temporal Attention-LSTM that additionally attends over pose/left-hand/right-hand landmark groups – under an identical, signer-independent evaluation protocol;

- An empirical demonstration that signer-independent splitting is not just a theoretical concern but measurably changes results on this dataset, and that scaling the training set from 500 to 1200 clips changes which architectural differences are visible above the noise floor;
- A critical analysis of *why* the proposed spatial attention extension underperforms the simpler temporal-only attention model in this low-resource regime, supported by confusion-matrix evidence, together with a multi-seed robustness check *across all three architectures* that quantifies how much a single training run can be trusted in this setting – and shows that the apparent single-run advantage of temporal attention over the baseline does not survive multi-seed re-evaluation.

The rest of this report is organized as follows. Section 2 reviews related work on recurrent sequence models, attention mechanisms, and sign language datasets. Section 3 describes the landmark extraction pipeline and the three model architectures. Section 4 describes the experimental protocol, presents the results, and discusses them critically. Section 5 concludes with the main findings and directions for future work.

2 Related Work

Recurrent sequence models. Long Short-Term Memory networks (LSTMs) [1] remain a standard baseline for sequence classification tasks such as ISLR: their gating mechanism lets them, in principle, retain information from early frames of a clip through to the final time step. In practice, however, a classifier that reads out only the final hidden state implicitly assumes that this compression is lossless, which becomes a weaker assumption as sequence length grows or as the informative signal is spread unevenly across time.

Attention mechanisms. Attention was originally introduced to let sequence-to-sequence models access every encoder time step instead of a single fixed-length summary vector [2], and was later generalized into the fully attention-based Transformer architecture [3]. In classification settings (rather than sequence-to-sequence generation), a simplified additive attention layer – a learned scalar score per time step, normalized with softmax, used to compute a weighted sum of hidden states – is a lightweight way to obtain the same benefit without adopting a full Transformer encoder. This project uses exactly this simplified form, and additionally extends it along a second axis: rather than only weighting *when* in the clip to look, a second attention module learns *where* on the body to look at each frame.

Landmark-based action and sign recognition. Estimating body and hand keypoints with a pretrained pose-estimation model, and classifying the resulting keypoint sequence, is a common way to make sign/action recognition robust to background and appearance variation [5]. This project follows that paradigm using MediaPipe’s Pose and Hand Landmarker models to obtain a 225-dimensional per-frame feature vector (33 pose points and 21 points per hand, each with x, y, z coordinates), which is the same representation size used in several skeleton-based sign/action recognition pipelines. A common further step in the skeleton-based

action recognition literature is to model the landmarks as a graph over anatomically connected joints and process them with a graph convolutional network (ST-GCN-style) rather than a flat per-frame vector; this project instead uses the flattened representation throughout, to keep all three LSTM-based architectures directly comparable under an identical input format, and treats the graph-based alternative as a natural direction for future work (Section 5).

Datasets and evaluation protocol. AUTSL [4] is a large-scale Turkish Sign Language dataset with 226 sign classes performed by multiple signers, explicitly designed to support signer-independent evaluation. Much of the isolated sign/action recognition literature evaluates with a random train/test split, which – as this project demonstrates empirically in Section 4 – can substantially overstate real-world performance whenever the same subject appears in both splits. This project follows AUTSL’s intended signer-independent protocol throughout. Unlike several other large-scale ISLR datasets recorded against a fixed or solid-colored studio background, AUTSL was deliberately collected across 20 different indoor and outdoor backgrounds, with varying lighting conditions and camera angles, to better reflect real-world recording conditions [4]. This background diversity is precisely the kind of nuisance variation that a landmark-based representation is designed to abstract away, which further motivates the choice of pipeline made in this project.

3 Methodology

Figure 1 shows the full pipeline: raw RGB video is converted into a per-frame landmark vector, augmented with motion features, and passed to one of three interchangeable classifier heads, all trained and evaluated under the same signer-independent split.

3.1 Landmark Extraction

Each video frame is processed with MediaPipe’s PoseLandmarker and HandLandmarker models [5] in video running mode. The resulting feature vector concatenates 33 pose landmarks and 21 landmarks per hand (left and right, disambiguated by MediaPipe’s handedness output), each with (x, y, z) coordinates, for a total of $33 \times 3 + 21 \times 3 + 21 \times 3 = 225$ dimensions per frame. If a hand or the pose is not detected in a given frame, the corresponding part of the vector is filled with zeros so that missing detections never crash the pipeline. Every video is then resampled to a fixed length of 30 frames (even-interval subsampling for longer clips, zero-padding for shorter ones), giving a fixed-size (30, 225) input tensor per sample. This target length was chosen based on AUTSL’s own reported clip statistics: the dataset’s average sample duration is approximately 2.05 seconds at 30 fps, i.e. roughly 61 frames per clip [4]; resampling to 30 frames therefore halves the average sequence length, keeping enough temporal resolution to preserve the overall motion trajectory (via even-interval subsampling rather than truncation) while keeping training and memory costs manageable for a single-GPU, small-dataset setting.

As an additional feature engineering step, an inter-frame velocity representation is computed as $v_t = x_t - x_{t-1}$ (with

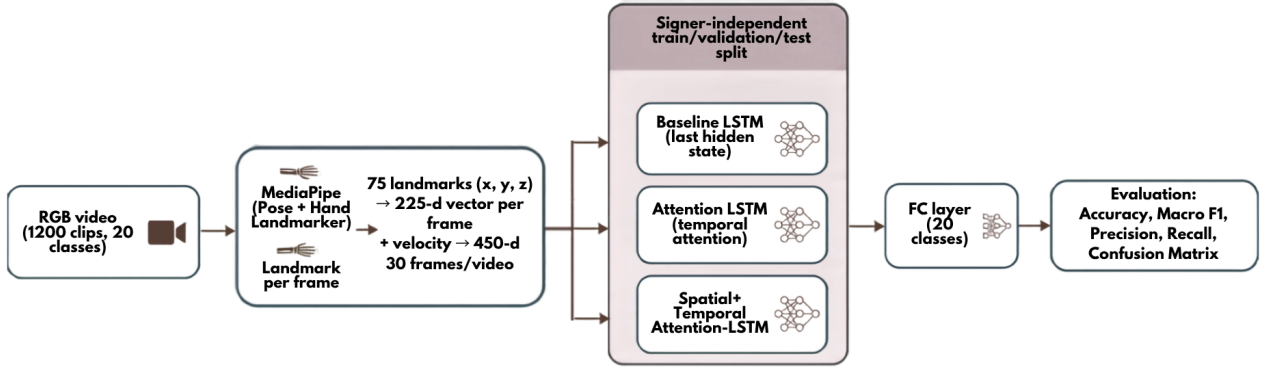


Figure 1: Overview of the pipeline. A raw video is converted into a 30-frame sequence of 450-dimensional landmark vectors (225 raw coordinates plus matching inter-frame velocity features) using MediaPipe’s Pose and Hand Landmarker models. The resulting sequence is fed to one of three classifier heads – Baseline LSTM, temporal Attention-LSTM, or the proposed Spatial+Temporal Attention-LSTM – each ending in a fully connected layer over 20 sign classes. All models are trained and evaluated under an identical signer-independent train/validation/test split.

$v_0 = 0$) and concatenated to the raw coordinates, doubling the per-frame feature size to 450. This gives every model explicit access to motion information – not just where a joint is, but how it is moving – rather than relying entirely on the recurrent model to infer velocity implicitly from consecutive positions. All results reported in this paper use this 450-dimensional representation.

3.2 Baseline LSTM

The baseline is a single-layer, single-directional LSTM (hidden size 64) whose classification head reads out only the final hidden state h_T , i.e. the network’s compressed summary of the entire 30-frame sequence, followed by dropout and a fully connected layer to 20 classes. This architecture assumes that the LSTM’s recurrent gating mechanism [1] successfully compresses all task-relevant information from every frame into this single vector.

3.3 Temporal Attention-LSTM

The Attention-LSTM keeps every time step’s hidden state $h_1, \dots, h_T$ instead of discarding all but the last one. A small linear layer maps each h_t to a scalar importance score, normalized across time with softmax to produce attention weights α_t , which are then used to compute a weighted sum $c = \sum_t \alpha_t h_t$ that replaces the final-hidden-state summary as input to the classification head. This is a simplified, classification-oriented form of additive attention [2], and lets the model give more weight to whichever frames happen to be most discriminative for a given sign, rather than assuming that information all propagates cleanly to the last time step.

Concretely, each hidden state is scored by a learned linear layer,

$$e_t = w_a^\top h_t + b_a, \quad t = 1, \dots, T, \quad (1)$$

and the scores are normalized across time with softmax,

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^T \exp(e_k)}, \quad (2)$$

to obtain the context vector

$$c = \sum_{t=1}^T \alpha_t h_t, \quad (3)$$

which replaces the final hidden state h_T as input to the classification head,

$$\hat{y} = \text{softmax}(W_c \text{Dropout}(c) + b_c). \quad (4)$$

3.4 Spatial+Temporal Attention-LSTM (proposed extension)

The proposed extension adds a second, complementary attention mechanism operating across body regions rather than across time. At every frame, the 450 landmark dimensions (225 raw coordinates plus 225 matching velocity features) are split into three groups – pose, left hand, right hand – based on their known index ranges in the feature vector. Each group is projected independently into a shared 32-dimensional embedding space via its own linear layer, scored with the same score $\rightarrow$ softmax pattern used for temporal attention (but normalized across the three groups instead of across time), and combined into a single fused, 32-dimensional per-frame representation via a weighted sum. This fused sequence is then processed by an LSTM exactly as in the temporal-attention model, with the same temporal attention layer applied to its outputs. The intent is to let the model learn not only *when* in a clip to focus, but also *which body part* is currently most informative – for example, favoring the hands over general body posture, which carries comparatively little sign-discriminating information for most of the vocabulary studied here.

Formally, at frame t , each landmark group $g \in \{\text{pose, left, right}\}$ with raw sub-vector $x_t^{(g)}$ is projected as

$$u_t^{(g)} = \text{ReLU}(W_g x_t^{(g)} + b_g), \quad u_t^{(g)} \in \mathbb{R}^{32}, \quad (5)$$

scored with a shared spatial scoring layer,

$$s_t^{(g)} = w_s^\top u_t^{(g)} + b_s, \quad (6)$$

and normalized across the three groups (not across time),

$$\beta_t^{(g)} = \frac{\exp(s_t^{(g)})}{\sum_{g' \in \{\text{pose}, \text{left}, \text{right}\}} \exp(s_t^{(g')})}, \quad (7)$$

to produce a fused, 32-dimensional per-frame representation

$$f_t = \sum_g \beta_t^{(g)} u_t^{(g)}. \quad (8)$$

The sequence $f_1, \dots, f_T$ is then processed by an LSTM, and the same temporal attention mechanism described in Eqs. (1)–(4) is applied to its hidden states to obtain the final context vector and prediction.

3.5 Training

All three models are trained with the Adam optimizer, cross-entropy loss, and identical hyperparameters (Table 2) to keep the comparison fair. To mitigate overfitting – observed in all three architectures on this small dataset – dropout is set to 0.5, and model selection uses the checkpoint with the highest validation accuracy across all 50 epochs (rather than the final epoch’s weights), since validation accuracy on a ~ 180 -example validation set is noisy enough that the last epoch is not necessarily representative.

4 Experimental Setup

4.1 Dataset

The dataset is a 20-word, class-balanced subset of AUTSL [4], selected from words with at least 40 available clips and capped at 60 clips per word, giving exactly 60 clips per class (1200 clips total, no class imbalance). Table 1 summarizes the dataset. Splitting uses `GroupShuffleSplit` grouped by signer ID (parsed from AUTSL’s `signerX_sampleY` filename convention), which guarantees that no signer appears in more than one of the train/validation/test sets – a signer-independent protocol, as opposed to a random split where the same signer’s clips could leak across splits. Because splitting is grouped by signer rather than by individual clip, the resulting split sizes are close to, but not exactly, the nominal 70/15/15% target; the test set that results from this procedure contains 246 examples.

Table 1: Dataset summary.

Property	Value
Dataset name	AUTSL (20-word subset)
Number of samples	1200
Number of classes	20 (60 per class)
Test samples	246
Train + validation samples	954 (signer-grouped)
Input format	30×450 -d landmarks

4.2 Training and Evaluation Protocol

Table 2 summarizes the training configuration, identical across all three models. Evaluation uses accuracy and

macro-averaged F1-score (to avoid any single class dominating the metric, even though classes are balanced here by construction), together with per-class confusion matrices for qualitative error analysis. As a robustness check, the best-performing model (Attention-LSTM) is additionally re-trained from scratch with three different random seeds, since a single run’s test accuracy can be misleading on a validation set this small.

Table 2: Training configuration.

Parameter	Value
Framework	PyTorch + MediaPipe
Optimizer	Adam
Learning rate	1×10^{-3}
Batch size	8
Epochs	50 (best-val-acc checkpoint)
Dropout	0.5
Hidden size	64
Loss function	Cross-entropy
Main metrics	Accuracy, macro-F1

4.3 Results

Table 3 reports test-set accuracy and macro-F1 for all models, both for a single training run and for a 3-seed mean $\pm$ standard deviation (seeds 42, 123, 7), the latter being the more trustworthy estimate given the run-to-run variance discussed below. Under single-run evaluation, the temporal Attention-LSTM outperforms both the baseline (+7.3 accuracy points) and a simple softmax-averaging ensemble of the baseline and attention models, which in this run reaches only 55.7% – lower than the Attention-LSTM alone, since the ensemble’s averaged prediction is pulled down by the weaker baseline. This picture changes further under 3-seed evaluation: the Baseline LSTM’s mean accuracy (51.6%) is in fact marginally *higher* than the Attention-LSTM’s (49.9%), and their standard-deviation ranges overlap almost entirely, meaning the two architectures cannot be reliably distinguished by mean accuracy alone at this dataset size. The Attention-LSTM’s one clear, consistent advantage is stability: its standard deviation (1.6) is more than twice as small as the Baseline’s (3.9), indicating a more predictable, less seed-sensitive model even though it does not reliably win on average accuracy. The Spatial+Temporal Attention-LSTM remains the weakest architecture under every evaluation protocol tested – single-run, ensemble, and 3-seed – with no overlap between its 3-seed range and either other model’s, confirming this is a robust rather than seed-dependent weakness.

Effect of dataset size. An earlier iteration of this project used a smaller subset of the same 20 words (25 clips/class, 500 clips total). At that scale, validation accuracy for all three architectures clustered within a few points of each other (~ 40 – 47%), with no architecture consistently ahead of the others across repeated runs – the differences were within the noise floor of such a small validation set. Only after scaling the training set to 60 clips/class (1200 total) did the Attention-LSTM’s single-run advantage over the baseline become clearly and consistently visible, while the spatial attention model remained the weakest architecture at both dataset sizes. As the 3-seed results above show, however, this single-

Table 3: Test-set comparison across models (246 test examples). Single-run values are from one training run per architecture; 3-seed values are mean $\pm$ std over three independent training runs (seeds 42, 123, 7) and are the more reliable estimate.

Method	Acc. (%)	F1 (%)
Baseline LSTM (1 run)	53.25	48.16
Baseline LSTM (3-seed)	51.63 $\pm$ 3.91	47.65 $\pm$ 3.80
Attention-LSTM (1 run)	60.57	59.37
Attention-LSTM (3-seed)	49.86 $\pm$ 1.64	44.99 $\pm$ 2.55
Spatial+Temporal Attn. (1 run)	33.74	30.13
Spatial+Temporal Attn. (3-seed)	37.67 $\pm$ 4.46	34.26 $\pm$ 4.32
Ensemble (Base+Attn., 1 run)	55.69	51.19

run advantage is itself within noise range once multiple seeds are considered – underscoring how easily a single favorable run can be mistaken for a stable architectural advantage, even after scaling up the dataset.

Effect of signer-independent splitting. During development, an early version of the splitting logic fell back to a random (non-signer-independent) split when signer IDs could not be parsed. Under that random split, validation accuracy was noticeably higher than under the signer-independent split used for all results reported here, for every architecture tested. This is consistent with signer leakage: a random split can place clips from the same signer in both the training and evaluation sets, letting a model partially recognize the *signer* rather than generalizing to the *sign*. All results in Table 3 use the stricter, signer-independent protocol.

Qualitative error analysis. Confusion matrices for the baseline, attention, and spatial models are shown in Figures 2–4. Two classes, *hafif* (“light”) and *hali* (“carpet”), are consistently among the hardest to classify: in the per-class breakdown, the Baseline LSTM scores F1 = 0.00 on *hali* and, despite perfect precision (1.00) on *hafif*, only 0.13 recall (F1 = 0.24) – meaning it rarely predicts *hafif* at all, but is never wrong when it does – while the Spatial+Temporal model scores F1 = 0.00 on *hafif* and 0.06 on *hali*. This is despite the dataset being perfectly class-balanced (60 examples per class), ruling out class imbalance as the explanation. This suggests these two signs are difficult to separate from others (in particular *calismak*, “to work”) in this landmark-based representation, rather than being under-represented in training data. The spatial attention model’s confusion matrix additionally shows a distinctive mode-collapse pattern: many unrelated classes (*baba*, *persembe*, *yemek*) are frequently misclassified as *turkiye*, visible as a dense vertical band in Figure 4, suggesting the model falls back on a small number of “default” classes when it cannot confidently discriminate.

4.4 Critical Discussion

Why attention helps – and where it does not. The temporal Attention-LSTM’s single-run improvement over the baseline (Figure 3 vs. Figure 2) is consistent with the motivating hypothesis that allowing the classifier to access every frame’s hidden state, rather than only the last one, helps when the disambiguating frame is not reliably the final one. However, the 3-seed evaluation complicates this picture: averaged over three independent runs, the Baseline LSTM is not reliably outperformed by the Attention-LSTM in mean accu-

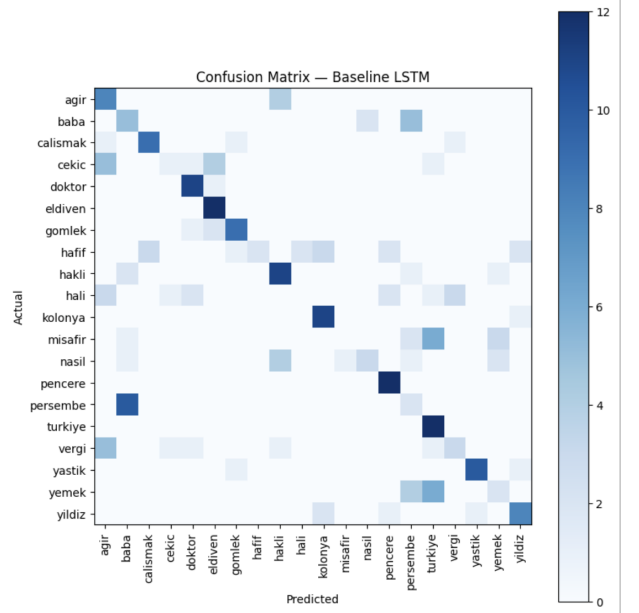


Figure 2: Confusion matrix, Baseline LSTM.

racy. What the multi-seed result does support is that attention improves *consistency*: its substantially lower standard deviation (1.6 vs. 3.9 accuracy points) suggests that weighting all time steps makes the model less sensitive to the specific random initialization and training trajectory, even if it does not guarantee a higher average score on a dataset this small. The confusion matrices in Figures 2–3 should therefore be read as illustrating one strong single run each, not the models’ typical behavior.

Why the proposed spatial attention extension does not help. The Spatial+Temporal Attention-LSTM underperforms both simpler models consistently, at both dataset scales tested, and under both single-run and 3-seed evaluation. Two factors plausibly explain this. First, the model introduces an information bottleneck: each landmark group (up to 198 raw+velocity dimensions for the pose group) is compressed into a 32-dimensional embedding *before* being seen by the LSTM, whereas the simpler Attention-LSTM lets the LSTM see the full, uncompressed per-frame vector. Second, the model has to learn two attention distributions jointly (over body regions and over time) instead of one, which is a harder optimization problem, and the training curves show a widening gap between (still-decreasing) training loss and (plateauing/noisy) validation loss consistent with overfitting on a dataset this size. The mode-collapse pattern observed in Figure 4 – with a large share of misclassifications funneled toward a small number of frequently-predicted default classes, most visibly *turkiye* – is consistent with a model that has not learned a reliable class-discriminative signal and instead falls back on a small set of frequently-correct default predictions. The 3-seed evaluation confirms this is not an artifact of a single unlucky run: the Spatial+Temporal model’s mean accuracy (37.7%) remains clearly below both other architectures’ 3-seed means, with non-overlapping standard-deviation ranges, indicating a robust rather than seed-dependent architectural weakness. This is a genuine negative result rather than an implementation error – the same score $\rightarrow$ softmax $\rightarrow$ weighted-sum pattern that

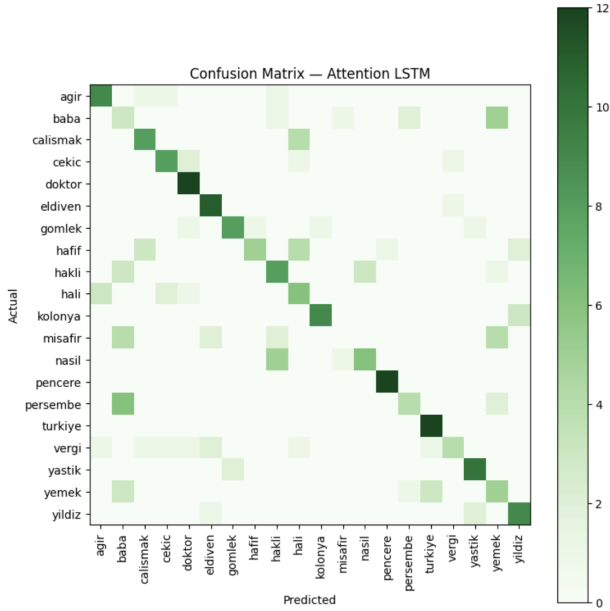


Figure 3: Confusion matrix, temporal Attention-LSTM.

works well for temporal attention does not straightforwardly transfer to the spatial axis in this low-resource regime, which is itself an informative finding about where added architectural complexity does and does not pay off with limited training data.

Run-to-run variance. With 3-seed results now available for all three architectures (Table 3), the gap between single-run and multi-seed performance is not specific to the Attention-LSTM: the Baseline’s best single run (53.3%) also differs from its 3-seed mean ($51.6\% \pm 3.9$), and the Attention-LSTM’s best single run (60.6%) sits well above its own 3-seed mean ($49.9\% \pm 1.6$) and outside one standard deviation of it. Two factors plausibly contribute to this. First, non-deterministic GPU kernels used by cuDNN’s LSTM implementation do not guarantee bit-identical results across runs even with `torch.manual_seed` fixed. Second, and likely more influential, is the model-selection protocol itself: each run selects the checkpoint with the highest validation accuracy across 50 epochs on a ~ 180 -example validation set, and picking the maximum of a noisy metric over many epochs is itself a form of selection bias that systematically inflates the reported number relative to the model’s typical performance. This is an important methodological point for any small-dataset, small-validation-set project of this kind: a single reported number, however strong, should not be treated as a stable estimate of a model’s true performance without a multi-seed check, and this project’s headline results are more honestly reported as the 3-seed means in Table 3 than as any single best run.

Limitations. The main limitations of this project are: (i) a relatively small dataset even after augmentation (60 examples/class), which likely explains both the overfitting observed across all architectures and the high run-to-run variance discussed above; (ii) two classes (*hafif*, *hali*) remain among the hardest to classify across models – for example, in the single-run results, the Spatial+Temporal model scores $F1 = 0.00$ on *hafif* and $F1 = 0.06$ on *hali*, and the Baseline scores $F1 = 0.00$ on *hali* despite non-trivial precision (though

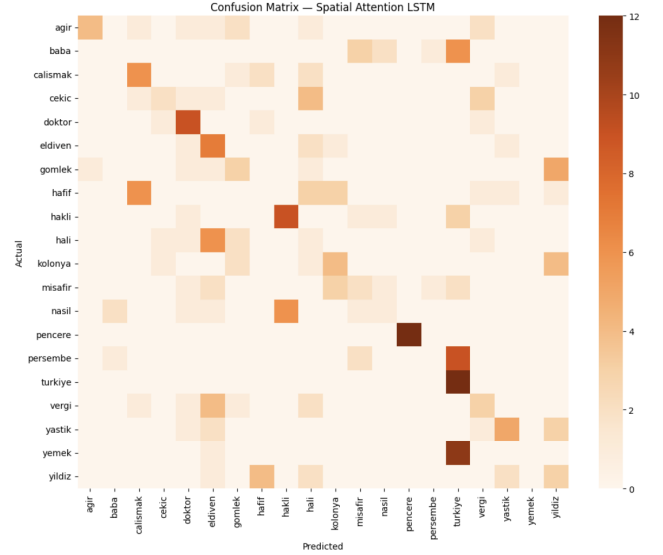


Figure 4: Confusion matrix, Spatial+Temporal Attention-LSTM. Note the dense vertical band at the *turkiye* column, indicating mode collapse toward a small set of classes.

not zero recall) on *hafif* – and this project does not go beyond confusion-matrix and classification-report evidence to explain why; (iii) absolute accuracy (~ 50 – 60% single-run, ~ 38 – 52% 3-seed mean) is well above chance (5% for 20 classes) but not yet at a level suitable for practical deployment.

5 Conclusion

This project built and compared three LSTM-based architectures for landmark-based isolated Turkish Sign Language recognition on a 20-word AUTSL subset, under a signer-independent evaluation protocol. In a single training run, a temporal Attention-LSTM outperformed a last-hidden-state baseline (60.6% vs. 53.3% test accuracy) and a two-model ensemble (55.7%); under 3-seed evaluation, however, this advantage did not hold – the Baseline’s 3-seed mean (51.6%) was marginally higher than the Attention-LSTM’s (49.9%), with heavily overlapping variance, though the Attention-LSTM was consistently more stable across seeds (roughly half the standard deviation). A novel Spatial+Temporal Attention-LSTM, extending attention to also select which body region is informative at each frame, was proposed and evaluated as this project’s main architectural contribution; it consistently underperformed the simpler temporal attention model under every evaluation protocol tested, and this report analyzes the likely causes (an information bottleneck and a harder joint optimization problem) using confusion-matrix evidence of mode collapse rather than simply reporting the negative result. Beyond architecture comparison, this project’s central methodological finding is that single-run comparisons between closely-matched architectures in this low-resource setting are unreliable and can reverse under multi-seed re-evaluation – alongside the complementary, consistently-replicated finding that signer-independent splitting measurably lowers (and makes more realistic) reported accuracy compared to a random split. Future work could in-

investigate why specific classes remain unrecognized, run additional seeds to narrow the confidence intervals in Table 3 further, and explore a graph-convolutional (ST-GCN-style) treatment of the landmark skeleton as a stronger alternative to the flattened per-frame vector used here.

Acknowledgments

This project uses the AUTSL dataset [4] and Google’s MediaPipe Pose/Hand Landmarker models [5] for landmark extraction.

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